Ateneo de Davao University
COMPUTER STUDIES CLUSTER
SCHOOL OF ARTS AND SCIENCES
P.O. Box 8016 Davao City
Philippines

Attestation of Primary Authorship* and AI Disclosure Statement

We, **Ralph Paolo Dulce and Yahyah Odin**, the proponents of thesis entitled **"A Coarse-to-Fine Semantic Conflict Detection System for Ex-Ante Davao City Ordinances Using Information Retrieval and Natural Language Inference"**, hereby declare that:

- We are the sole authors of this thesis and its technical output.
- We have fully acknowledged and referenced the ideas, techniques and work of others, whether published or unpublished, in our thesis.
- We have prepared the thesis specifically for the degree of **Bachelor of Science in Computer Science** while under supervision at Ateneo de Davao University.
- Our thesis document does not contain work extracted from a thesis, dissertation or research paper previously presented for another degree or diploma at this or any other university.
- Our technical output was not subcontracted to a third party.

AI Disclosure Statement

- We hereby declare that any use of artificial intelligence (AI) tools in the preparation of this thesis and its technical output:

AI Tool Used	Purpose and Description of Use	Sections/Components Where AI Was Used
Google Antigravity / Gemini Models	Assisted as an agentic pair-programmer for Python engineering: scaffolding modular codebase architecture (`src/`), automated execution harnesses, data pipelines for dual-corpus JSONL compilation (27,357 enactments, 178,494 provisions), batch OCR wrappers, and engineering the offline interactive web prototype interface (`demo/`).	Codebase Pipelines (`src/`, `scripts/`) and Interactive Web Prototype (`demo/`)
Google Antigravity / Gemini Models	Assisted in LaTeX typesetting, multi-pass compilation orchestration (`pdflatex/bibtex` in `scripts/build_paper.py`), resolution of build diagnostics, formatting dedicated full-page landscape plates (`pdflscape`) for empirical tables, managing front-matter listings, and synchronizing the automated single-source-of-truth numbers registry.	Thesis Manuscript (`main.tex`, Chapters 1–5), Appendices A–R, and Numbers Registry
Gemini Flash 3.8 (Google)	Assisted in syntactic clause drafting and formal legislative enacting phrase templating of candidate local ordinance provisions under strictly human-specified statutory premises, legal mechanisms, and conflict types (Section 3.5.3), conforming strictly to Davao City Sangguniang Panlungsod and DILG drafting manuals without fabricating legal facts.	Methodology (Section 3.5.3) and Ground Truth Benchmark Formulation (`data/`)

**Adapted from Faculty of Graduate Research, Statement of Authorship and Thesis Length for PhD Students, Victoria University of Wellington, 2009*

AI Tool Used	Purpose and Description of Use	Sections/Components Where AI Was Used
Claude 3.5 Sonnet (Anthropic)	Assisted as an adversarial simulated peer reviewer and mock defense interrogator: stress-testing scientific claims, identifying narrative overclaiming, and conducting forensic checks on mathematical and statistical consistency (McNemar 2×2 contingency tables on $N = 53$, Friedman ANOVA ranks, Wilson score CIs, and Bayesian low-prevalence screening dynamics).	Thesis Manuscript (Chapters 1–5, Abstract, Conclusions, and Statistical Disclosures)
Claude 3.5 Sonnet / Gemini Models	Assisted in editorial prose refinement, sentence flow transitions, eliminating repetitive phrasing or conversational filler, and ensuring adherence to academic continuous paragraph formatting in strict accordance with departmental writing guidelines.	Manuscript Prose Polish across Chapters 1–5, Abstract, and Appendices
Evaluated ML Models (Research Subjects)	Deep neural cross-encoders (`DeBERTa-v3`, `ModernBERT`, `Legal-BERT`, `all-MiniLM-L6-v2`) and vision-language models (`Qwen2.5-VL-3B-Instruct`) were evaluated strictly as offline empirical machine learning models and experimental research subjects under deterministic evaluation on local LGU hardware, not as generative text authors.	Methodology and Experimental Results (Chapters 3 & 4, Appendices A–R)

We further declare that:

- We have accurately disclosed the use of AI tools in the preparation of this thesis and its technical output.
- We take full responsibility for the accuracy, originality, integrity, and academic quality of the submitted work, including any content generated or assisted by AI tools.
- We have reviewed and verified all AI-generated or AI-assisted content incorporated into the thesis and its technical output and have ensured that appropriate acknowledgements and citations are provided, where necessary.

By signing below, we affirm that the statements above are true and accurate.

Signed: _____ **Date:** _____
Ralph Paolo Dulce

Signed: _____ **Date:** _____
Yahyah Odin

**Adapted from Faculty of Graduate Research, Statement of Authorship and Thesis Length for PhD Students, Victoria University of Wellington, 2009*